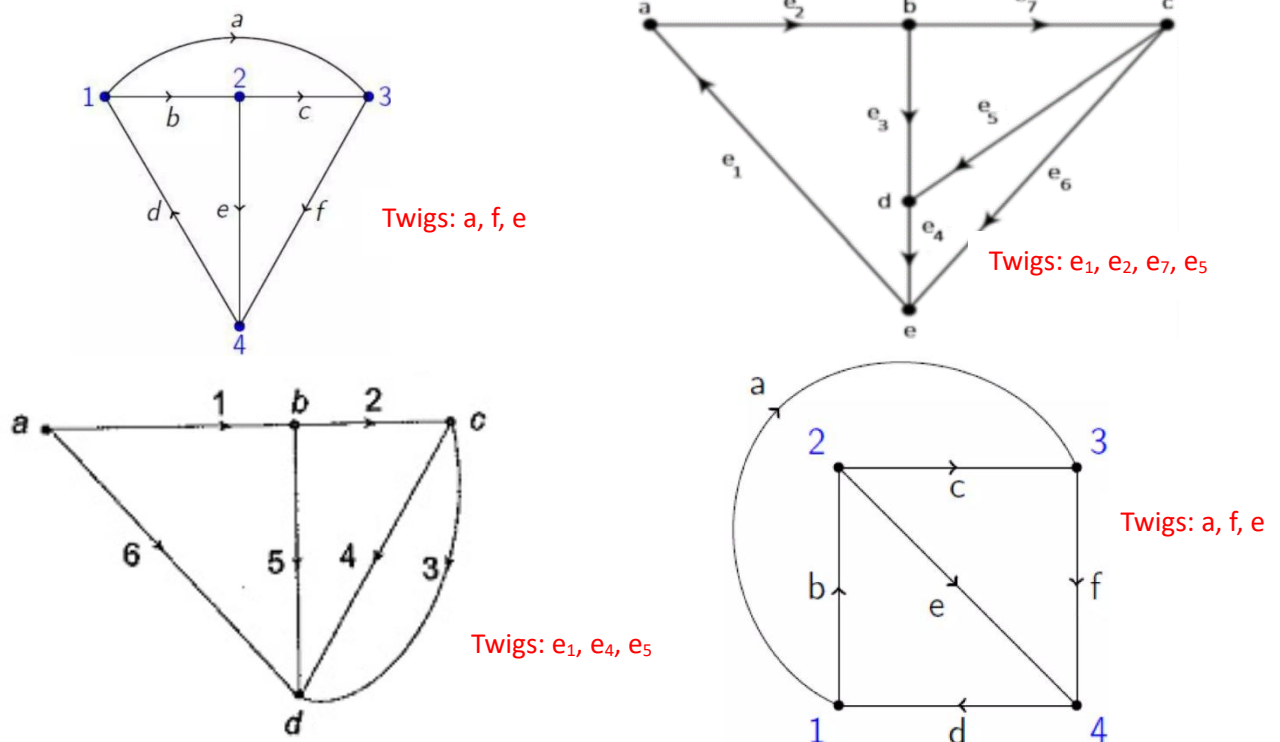


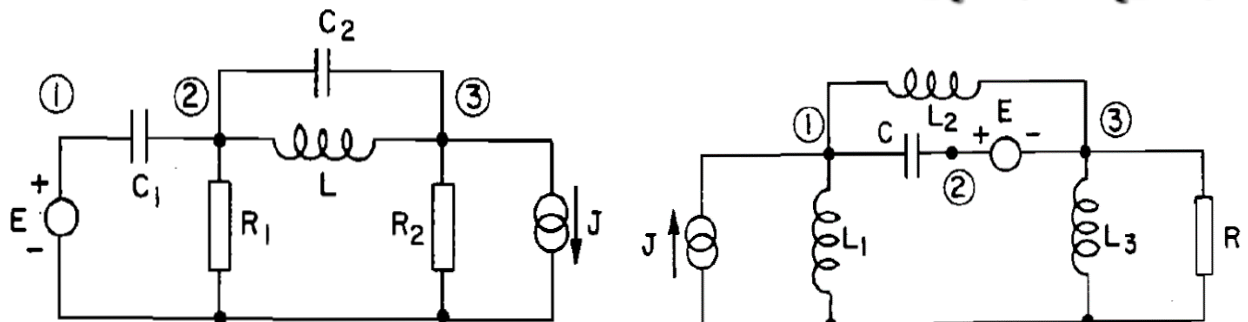
Tutorial-3

- For each of the oriented graphs shown in figures below, find the incidence matrix A, f-cutset matrix Q and f-circuit matrix B.



- Draw the oriented graphs for the networks shown in figures below. Show the validity of the orthogonality relations by using the Q and B matrices.

$$BQ^T = 0 \text{ or } QB^T = 0$$



- Transform the voltage sources in the above figures into equivalent current sources, draw appropriate graphs, and apply the topological formulation of nodal equations to these networks (Incidence matrix based).
- Repeat the above question using fundamental cutset matrix. $QY_bQ^T V_t = -QJ_b$
- Transform the current sources in the above figures into equivalent voltage sources, draw appropriate graphs, and apply the topological formulation of loop equations to these networks.